

GETTING SPACE RIGHT: AGENTIC REWARDS FOR IMAGE GENERATION

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ABSTRACT

Reinforcement learning (RL) is promising for spatial text-to-image generation, but existing work largely focuses on policy optimization, assuming accurate rewards and effective online rollouts. We observe in-domain gains with out-of-domain degradation, exposing two limitations: unreliable spatial rewards in open-world settings and rollouts lacking basic-to-complex progression. We propose a curriculum-guided RL framework. First, we introduce SPATIALCRITIC, an agentic reward model equipped with a spatial perception harness that combines reusable verification guidance with specialized vision tools. We then construct spatial preference datasets through targeted counterfactual perturbations and perform agent-harness alignment to improve tool selection and evidence integration efficiently. Second, curriculum learning organizes shared foundational capabilities and their compositional dependencies in a capability graph for mastery-guided rollouts, periodic review, and targeted remediation. On SpaRW-Eval, SPATIALCRITIC achieves 95.1% verification accuracy, outperforming 320B GLM-5.3-Flash with a 40× smaller 8B backbone and raising average baseline scores from 78.4 to 83.9 on SD3.5-Medium and 83.2 to 86.5 on BAGEL. See the [project page](#).

1 INTRODUCTION

Enhancing spatial intelligent in text-to-image (T2I) generation, i.e., generating accurate spatial relationships faithful to text prompts (Ghosh et al., 2023; Wang et al., 2026c), remains a critical and challenging task in T2I post-training. Recently, reinforcement learning (RL) has emerged as a prevailing paradigm for aligning pre-trained models with task-specific goals.

RL aligns a T2I model by generating images from training prompts, evaluating rewards, and updating the policy. Thus, its effectiveness depends on three aspects: (1) rollout organization, (2) reward accuracy, and (3) policy optimization. Existing work mainly advances the third (Black et al., 2024; Clark et al., 2024; Liu et al., 2025a; Zheng et al., 2026), often assuming that rollouts support effective learning and rewards provide reliable feedback. This leaves two questions: (*Q1*) Are reward models accurate and robust enough in open-world scenarios? (*Q2*) Can rollout organization support progressive learning from basic spatial capabilities to complex spatial compositions?

Unfortunately, these assumptions fail in spatial text-to-image generation: RL-aligned models improve steadily on in-domain (ID) benchmarks but *generalize poorly out of domain (OOD)*, eventually falling below the pre-trained model (Fig. 1). This failure stems from repurposing benchmarks such as GenEval as training prompts and their rule-based scoring pipelines as on-line rewards, exposing two critical flaws: (1) inaccurate and non-robust rewards, as benchmark-specific scoring generalizes poorly to open-world scenarios, especially under object occlusion (Tang et al., 2026); and (2) rollouts targeting complex compositions without progressive learning. Reused benchmark prompts emphasize joint constraint satisfaction, neglecting fundamental spatial capabilities and their progressive composition into complex ones.

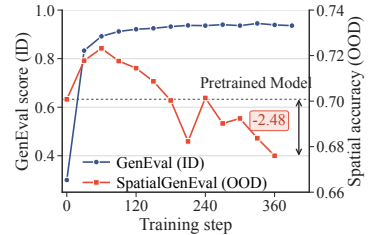


Figure 1: **Generalization collapse.** ID accuracy improves while OOD performance falls.

To overcome the brittleness of rule-based rewards, recent efforts pursue learned reward modeling across two paradigms: *regression-based models* that map dual-encoder or VLM representations

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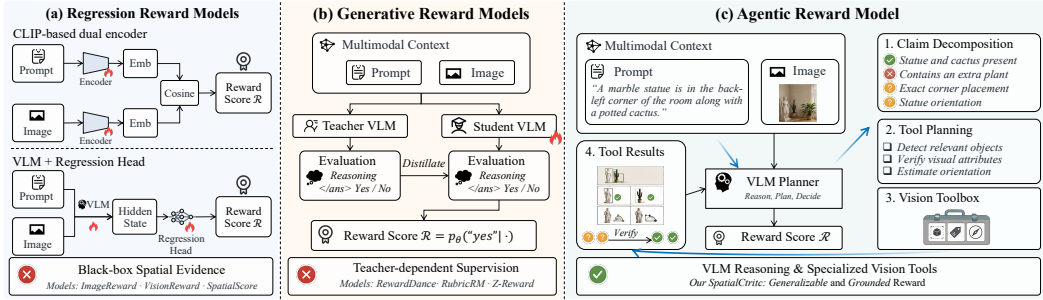


Figure 2: **Comparison of reward-modeling paradigms.** (a) Regression-based scoring; (b) teacher-guided generative evaluation; and (c) our agentic reward model, SPATIALCRITIC.

into scalar preference scores (Xu et al., 2026a; Tang et al., 2026), and *generative models* that emulate deliberative evaluation via reasoning rationales or dynamic rubrics (Wang et al., 2026a; Kan et al., 2026), as shown in Fig. 2. However, neither ensures reliable spatial verification. Regression models treat foundation models as black-box feature extractors, underutilizing native reasoning and geometric depth priors. Meanwhile, generative models fundamentally rely on ungrounded teacher generation, and struggle to explicitly and reliably resolve geometric quantities such as occlusion and orientation. Beyond reward modeling, progressive capability learning remains largely unexplored. *These gaps motivate us to combine a reward model leveraging vision foundation models for verification with a progressive curriculum, jointly ensuring accuracy and generalizability.*

Contributions. We propose a curriculum-guided RL paradigm for spatial T2I post-training, comprising SPATIALCRITIC, an agentic reward model that leverages vision foundation models for robust supervision, and a progressive curriculum from basic spatial capabilities to complex compositions.

First, we design an agentic reward model, SPATIALCRITIC, that harnesses specialized vision foundation models for grounded spatial verification, optimized via Reinforcement Learning with Verifiable Rewards (RLVR). Its development comprises three components: (1) *Agent with a spatial perception harness.* To enhance the agent’s spatial perception capabilities, we design an agent equipped with a spatial harness that provides reusable verification guidance and executes specialized perceptual tools. Through this harness, the agent decomposes prompts into atomic claims and selectively queries specialized tools under uncertainty. (2) *Spatial preference dataset construction.* To enhance the agentic ability on the harness, the verifiable RL environment is necessary for post-training. We therefore construct a spatial preference dataset by counterfactually perturbing targeted spatial relations, such as occlusion, depth, and orientation, while keeping other visual content invariant. (3) *Agent–harness alignment via RLVR.* Building upon the agent and the preference dataset, we optimize the agent’s harness ability, such as tool-selection and evidence-integration, using verifiable outcome rewards and token-budget penalties to improve spatial judgment accuracy and efficiency.

Second, we introduce curriculum learning for spatial T2I post-training, which organizes policy roll-outs from basic spatial capabilities to complex compositions rather than directly optimizing over an unstructured mixture of complex tasks. Our key insight is that *higher-order spatial tasks involve different combinations of shared foundational capabilities.* For example, multi-object layout combines counting, and topology, while occlusion combines depth and topology, sharing topology capabilities. Accordingly, we introduce a Compositional Capability Graph (CCG) to capture these dependencies. Building on CCG, capability mastery guides progression, while periodic review identifies compositional weaknesses for targeted remediation. Together, these designs enable feedback-driven progression from basic capabilities to complex compositions.

Finally, *comprehensive experiments* validate accurate preference judgments, effective reward-guided image selection, and improved RL alignment. SPATIALCRITIC scores 95.1% on SpaRW-Eval. Curriculum RL raises baseline averages to 83.9 on SD3.5-Medium and 86.5 on BAGEL.

2 RELATED WORK

Reward Models for Visual Generation. Visual reward modeling has evolved from feature-similarity heuristics (Radford et al., 2021) to learned human preference representations. Early

approaches adopted dual encoders (Xu et al., 2023; Kirstain et al., 2023; Wu et al., 2023; Zhang et al., 2024) to predict scalar preference scores via Bradley-Terry objectives. To overcome single-scalar bottlenecks, subsequent paradigms deployed multimodal VLMs to predict disentangled rewards across aesthetics, alignment, and visual artifacts (Xu et al., 2026a; Liu et al., 2025b; Ma et al., 2025; Liu et al., 2026a; Tang et al., 2026). More recently, generative reward models emit dynamic rubrics, critique rationales, preference tokens, or calibrated score distributions (Wu et al., 2025b; Jin et al., 2026; Kan et al., 2026; Wang et al., 2026b;a). Closest to our work, SpatialReward (Zhou et al., 2026b) grounds decomposed spatial constraints with expert detectors and reasons over the resulting observations. Its verification procedure is nonetheless determined prior to observation, following a pre-defined routine per task category (Zhou et al., 2026a). We instead learn evidence acquisition as a per-claim decision over an open tool catalog, aligned to the harness under outcome supervision.

Spatial Intelligent Training Paradigm. Recent spatial intelligence studies reveal that learning complex spatial reasoning without grounded perceptual foundations induces severe shortcuts and optimization instability (Li et al., 2025; Liang et al., 2026; Yang et al., 2026). Consequently, structured spatial data and staged learning are explored in multimodal representation (Cai et al., 2026; Li et al., 2025; Liang et al., 2026) and embodied planning (Yang et al., 2026; Qian et al., 2026). However, these works primarily target spatial understanding and embodied planning through data scaling, representation adaptation, or staged alignment. In contrast, our work centers on spatial T2I post-training, where a capability dependency graph and an absolute mastery probe guide online prompt scheduling, periodic review, and targeted remediation.

3 SPATIALCRITIC: TOOL-MEDIATED AGENTIC REWARD MODEL

3.1 AGENT WITH A SPATIAL PERCEPTION HARNESS

Problem Formulation. For spatial intelligence in text-to-image generation, the reward model requires to assess the spatial consistency of the generated image with the spatial constraints in the prompt, including object existence, attribute binding, counting, comparison, 2D position, background placement, distance, depth, orientation, topology, occlusion, multi-object layout, and 3D pose. Action, physical-plausibility, and image-quality checks provide auxiliary context. Preference judgments typically follow either pairwise comparison or pointwise scoring $S(\mathbf{I}, c)$ (Xu et al., 2023; Wang et al., 2025c). Our framework supports both, but primarily uses pairwise comparison. Given $\mathbf{x} = (\mathbf{I}_A, \mathbf{I}_B, c)$, SPATIALCRITIC predicts which image better satisfies the spatial constraints in c , yielding $\mathbf{I}_A \succ \mathbf{I}_B$. For score-wise setting, the preference is predicted by $S(\mathbf{I}_A, c) > S(\mathbf{I}_B, c)$.

Motivation: From Complex Spatial Relations to Atomic Verification. Spatial reward assessment benefits from complementary semantic and geometric capabilities. While VLMs excel at open-ended prompt comprehension, they fundamentally struggle with fine-grained spatial relations such as depth ordering and occlusion boundaries (Liu et al., 2026b). Conversely, fixed detector-based evaluators (Ghosh et al., 2023) rely on rigid and predefined heuristics and fail on complex open-world spatial relations. For example, consider the prompt “A palm tree is closer to the camera than a blue beach umbrella”. The VLM is hard to accurately locate the position of the tree and the umbrella, and estimates their depth ordering, while the detector-based evaluator can provide the precise information. In this case, the complex spatial relation is decomposed into three atomic verification tasks: checking the presence of both objects, the umbrella’s color, and their depth ordering.

Key Challenges. As discussed above, spatial reward modeling benefits from combining semantic reasoning with specialized perceptual evidence. Beyond integrating tool outputs within a predefined verification pipeline (Zhou et al., 2026b), we seek an agent that dynamically orchestrate vision foundation models for open-world spatial verification. This raises two complementary challenges: (1) *for high-frequency and canonical spatial relations*, how to make recurring verification patterns reliable and reusable? (2) *for complex and unseen spatial relations*, how to adapt these patterns to unfamiliar spatial compositions without being restricted to predefined tools pipeline or relying solely on unverified VLM judgments? Fundamentally, this requires balancing standardized execution on common and frequent relations with compositional flexibility on unseen and novel ones.

Spatial Perception Harness. The harness integrates a modular *perceptual toolkit*, covering grounding, counting, segmentation, depth, 3D geometry, orientation, and contextual checks, with structured *skill guidance*, specifying evidence requirements and output interpretations. For claims flagged as

uncertain during decomposition, the harness guides tool planning along two paths. (1) When a claim matches a *high-frequency relation*, the agent uses a *standardized skill* with a fixed local tool routine and evidence interpretation, addressing the first challenge. (2) Otherwise, for *complex or unfamiliar relations* requiring query-specific composition, the harness provides an open tool catalog with capability descriptions, guiding the agent to *compose and sequence targeted tool plans*. In Fig. 3(b), the plan grounds the instances, checks each dog’s orientation relative to the fountain, and aggregates these relations to verify the requested counts. This balances robust execution on high-frequency relations with flexible open-world composition. Toolkit details appear in App. C.

Agent Pipeline. As shown in Fig. 2(c), SPATIALCRITIC verifies spatial consistency through four interactive stages: (1) *Claim decomposition*: the VLM policy decomposes the prompt into atomic spatial claims and flags perceptual uncertainties across candidate images; (2) *Tool planning*: guided by the spatial perception harness, the agent identifies missing visual evidence and plans targeted tool calls; (3) *Tool execution*: the harness invokes the selected vision foundation models and collects structured perceptual observations; and (4) *Evidence integration*: the policy integrates these observations into its reasoning to resolve outstanding claims and produce the final preference verdict.

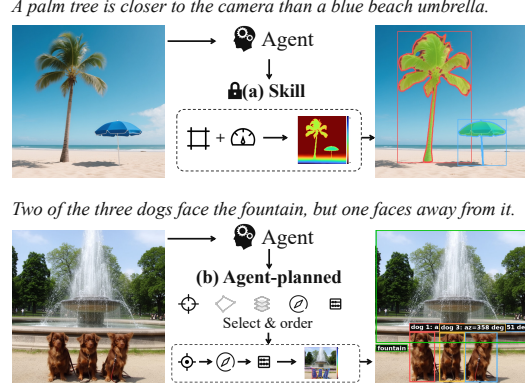


Figure 3: **Spatial verification.** (a) Fixed mask–depth routine; (b) agent-planned tool use.

3.2 VERIFIABLE SPATIAL PREFERENCE DATASET

Although the proposed spatial perception harness balances standardized execution for frequent tasks with flexibility for out-of-distribution cases, efficiency and accuracy remain limited by redundant or failed tool calls, conflicting tool outputs, and ambiguity among similar tools. Fundamentally, these issues reflect a *mismatch between the agent’s policy and the spatial perception harness*, requiring the agent to adapt its invocation, selection, and judgment policies to the harness.

This mismatch is difficult to resolve through harness design alone, as agent and tool capabilities vary across tasks and are difficult to align a priori. We therefore employ RLVR to learn adaptive tool selection and evidence integration through interaction, balancing verification accuracy and tool-use cost without predefined execution trajectories. To this end, we construct a spatial preference dataset of image pairs with explicit preference labels for judgment verification and reward computation.

Goal. To make these preferences effective reward signals for RL, the dataset should satisfy two requirements: *diversity* across spatial tasks and visual contexts, supporting generalizable use of the harness; and *accuracy* of the preference labels, providing reliable feedback for learning tool selection and evidence integration. We address diversity through *diverse task and scene coverage*, and accuracy through *constraint-targeted preference construction*, which combines explicit prompt constraints with text- and image-level validation.

Diverse task and scene coverage. Our construction covers a registry of 36 *scene settings* and a taxonomy of 13 *task categories*, with 3D pose held out from agent training. We balance target-category coverage and sample compatible scenes, supplying their objects, scales, and reference-frame rules to the LLM. Duplicate checks and multiple image generators further broaden textual and visual variation. App. B.2 lists the task and scene inventories, the complete generation system prompt, and the task-conditioned input fields.

Counterfactual preference construction & data pipeline. To obtain reliable spatial preference labels, we define unambiguous constraints and verify that the generated images form a valid contrast. We construct image preferences from reference prompts $\mathcal{D}_0 = \{c\}$ in four stages as shown in Fig. 4: (1) *Prompt-pair construction*. Targeted perturbations provide compositional negatives and counterfactual image pairs (Ma et al., 2023; Tang et al., 2026). Conditioned on the target category and scene constraints, DeepSeek-V4-Flash (DeepSeek-AI, 2026) jointly generates an unambiguous reference c^+ and three alternatives c^- , forming $\mathcal{D}_1 = \{(c^+, c^-)\}$ (Fig. 4(a)). Each alternative is intended to violate one constraint while preserving the others; at least one targets the designated

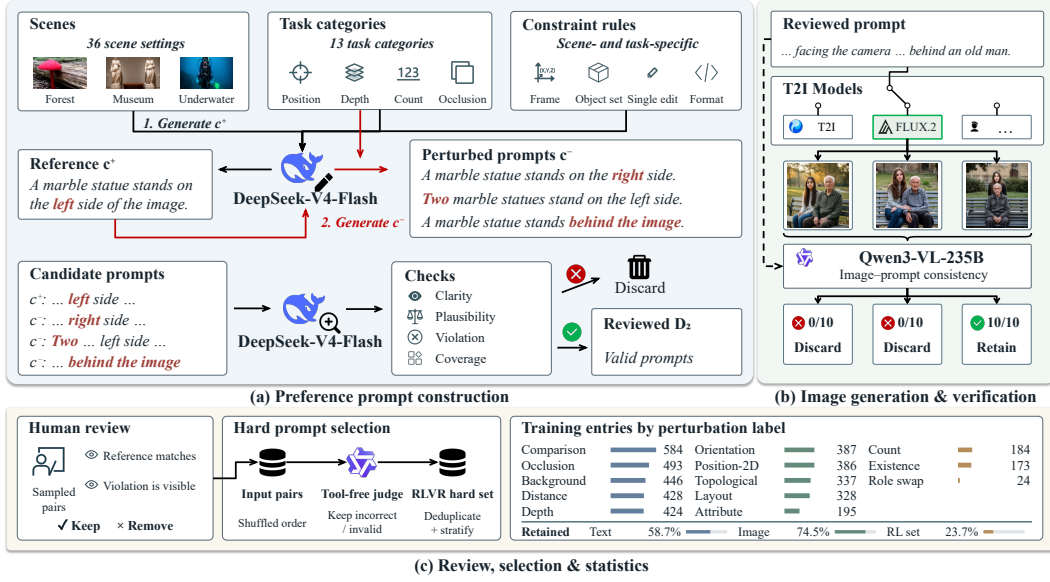


Figure 4: **Spatial preference dataset construction.** (a) Conditioned prompt generation, perturbation, and review. (b) Multiple image generation and VLM verification. (c) Human review, hard-set selection, and dataset statistics.

category. The generation instructions are provided in App. B.2. (2) *Text-level validation*. A separate review call to DeepSeek-V4-Flash yields $\mathcal{D}_2 = \text{Filter}(\mathcal{D}_1)$ by filtering pairs for clarity, physical plausibility, and explicit violations, assigning perturbation categories, and checking target-category coverage. (3) *Image-level verification*. As illustrated in Fig. 4(b), we render both prompts of each pair in $\mathcal{D}_2$ with FLUX.2-dev (Black Forest Labs, 2025), Hunyuan-Image-3.0 (Tencent Hunyuan Foundation Model Team, 2025), and Qwen-Image (Wu et al., 2025a), using the same generator within a pair, and retain only pairs that pass a Qwen-VL-235B (Bai et al., 2025a) check of reference compliance and the intended violation, so that preferences reflect image content rather than textual changes alone. We apply best-of- n selection over the rendered candidates and keep the highest-scoring reference-compliant image as $\mathbf{I}^+$. (4) *Human inspection*. To mitigate automated verification noise, human annotators independently audit candidate pairs retained from image-level verification against identical criteria, confirming that $\mathbf{I}^+$ satisfies c^+ and $\mathbf{I}^-$ exhibits the intended violation; failing pairs are removed. The counterfactual prompt c^- is discarded after verification, yielding $\mathcal{D}_3 = \{(c^+, \mathbf{I}^+, \mathbf{I}^-)\}$, where $\mathbf{I}^+ \succ \mathbf{I}^-$ under c^+ . The inspection protocol and quality statistics appear in App. B.1.

3.3 AGENT-HARNESS ALIGNMENT VIA RLVR

Building on the spatial perception harness and the spatial preference dataset, we align the agent’s tool-selection and evidence-integration policy with the harness through RLVR. We optimize the agent policy while keeping the harness and perceptual tools fixed, using verifiable outcome rewards and token-budget penalties to improve spatial judgment accuracy and token efficiency.

Hard set selection. The initial verification capability of the agent can induce an imbalance between positive and negative outcome rewards. To provide balanced feedback across spatial tasks, we identify hard where the tool-free base model fails. Combining this difficulty mining with category rebalancing prioritizes complex spatial relations over elementary recognition. Deduplication then yields a balanced training set as shown in Fig. 4(c), with complete protocols and counts in App. B.1. Based on this dataset, we define rewards for preference judgment and tool use.

Reward formulation. To improve token efficiency while preserving judgment accuracy, we combine an outcome reward with a token-budget penalty conditioned on correct judgments:

$$\mathcal{R}(\tau) = \mathcal{R}_{\text{out}}(\tau) + \mathbb{I}[\hat{y}(\tau) = y^{\text{gt}}] \text{clip}(\mathcal{R}_{\text{tok}}(\tau), -\kappa, 0). \quad (1)$$

where $\hat{y}(\tau)$ is the predicted preference, y^{gt} is its ground-truth label, and κ bounds the auxiliary penalty. We set $\mathcal{R}_{\text{out}}$ to +1, -1, and -1.5 for correct, incorrect, and invalid results, respectively.

The token penalty is $\mathcal{R}_{\text{tok}}(\tau) = -\alpha \max(0, n_{\text{tok}} - n_0)$, where n_{tok} counts agent-generated tokens, excluding input prompts and tool observations; n_0 is the free token budget and α penalizes each excess token. This objective encourages accurate judgments, targeted tool use, and concise evidence integration with fewer tokens and lower latency, as illustrated in Figs. 6, 7, and S1.

Policy optimization. With the dataset and reward, we optimize π_ϕ using GRPO (Shao et al., 2024). For each randomly ordered image pair, we sample G rollouts under $\pi_{\phi_{\text{old}}}$ following Sec. 3.1, then update the policy using group-relative reward advantages. Details appear in App. D.1.

4 CURRICULUM LEARNING FOR SPATIAL T2I POST-TRAINING

Building on SPATIALCRITIC, we introduce curriculum learning to address the lack of basic-to-complex progression. This requires (1) acquiring foundational spatial capabilities; and (2) composing them for combination. We use a *compositional capability graph* to define these capabilities and their dependencies, and a *progressive learning strategy* to schedule foundational and compositional training accordingly, guided by policy competence, review, and remediation.

4.1 COMPOSITIONAL CAPABILITY GRAPH

To guide foundational learning and composition, we examine how spatial capabilities relate. Our key insight is that *higher-order spatial tasks combine shared foundational capabilities in different ways*. CCG represents these foundations and dependencies beyond a single easy-to-hard ordering. Specifically, it is a directed acyclic graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with 13 capability nodes, where each edge (u, v) requires u to precede v in training. In Fig. 5, Layout and Occlusion share Topology; Layout additionally depends on Count and Position, and Occlusion on Depth. These prescribed dependencies provide an inductive bias for curriculum design, with the complete mapping in App. D.4. This graph facilitates training scheduling with a progressive learning strategy.

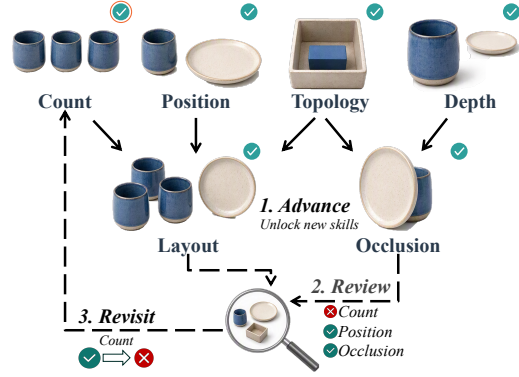


Figure 5: **CCG-guided progressive learning.** Solid arrows show capability dependencies.

4.2 PROGRESSIVE LEARNING WITH REVIEW AND REMEDIATION

Our progressive learning framework has two components in Fig. 5: (1) *forward progression*, which develops foundational capabilities and advances to dependent tasks based on mastery; and (2) *review and remediation*, which addresses forgetting and joint-constraint failures by reassessing learned capabilities individually and jointly.

Forward progression. Let $\mathcal{P}(v)$ denote the prerequisites of capability v in CCG. At assessment round t , we obtain the mean evaluation reward $\mathcal{S}_v^t$ for each assessed capability. To reduce sensitivity to outliers, we update its running score as $\bar{\mathcal{S}}_v^t = \text{Smooth}(\mathcal{S}_v^t, \bar{\mathcal{S}}_v^{t-1})$, where Smooth is an exponential moving average, detailed in App. D.4. Mastery requires $\bar{\mathcal{S}}_v \geq \tau$ and at least $N_{\min}$ image assessments. To prevent curriculum deadlock, unmastered capabilities permit fallback progression after $N_{\max}$ image assessments. We define the mastered and fallback sets as

$$\mathcal{M} = \underbrace{\{v \in \mathcal{V} \mid \bar{\mathcal{S}}_v \geq \tau, n_v \geq N_{\min}\}}_{\text{Mastered capabilities}}, \quad \mathcal{F} = \underbrace{\{v \in \mathcal{V} \mid v \notin \mathcal{M}, n_v \geq N_{\max}\}}_{\text{Unmastered at assessment limit}}. \quad (2)$$

where n_v counts image assessments for capability v . $\mathcal{M}$ contains mastered capabilities, while $\mathcal{F}$ contains those still unmastered at the assessment limit. Both sets permit downstream progression, but capabilities in $\mathcal{F}$ remain candidates for review and remediation. A successor capability v becomes *unlocked*, or eligible for training, when all its prerequisites belong to either set:

$$\text{Unlocked}(v) \iff \mathcal{P}(v) \subseteq \mathcal{M} \cup \mathcal{F}. \quad (3)$$

Fig. 5 illustrates progression from foundational to composite capabilities. Once Count, Position, Topology, and Depth meet the mastery criterion, they enter $\mathcal{M}$. Layout requires Count, Position, and

Topology, while Occlusion requires Topology and Depth. Both sets therefore lie in $\mathcal{M}$, unlocking Layout and Occlusion for training. The *Advance* phase shifts training toward these capabilities, with policy updates on the selected prompts using standard diffusion RL (Liu et al., 2025a).

Review and remediation. Forward progression alone does not ensure capability retention or compositional performance. (1) *Review* periodically evaluates capabilities in $\mathcal{M} \cup \mathcal{F}$ through individual and compositional tasks, updating $\mathcal{S}_v^t$ using the same evaluation and smoothing procedure. Capabilities failing the mastery criterion in Eq. (2) form the remediation set $\mathcal{C}$ and are removed from $\mathcal{M}$ via $\mathcal{M} \leftarrow \mathcal{M} \setminus \mathcal{C}$. (2) *Remediation* applies targeted training to $\mathcal{C}$ before forward progression resumes. Capabilities satisfying the mastery criterion in either stage enter $\mathcal{M}$ and leave $\mathcal{F}$; those remaining in $\mathcal{F}$ remain eligible for subsequent review and remediation. For example, jointly assessing Count, Position, and Occlusion in Fig. 5 yields $\mathcal{C} = \{\text{Count}\}$ when only Count fails the mastery criterion. The *Revisit* step then applies counting-specific training to improve performance under joint constraints. App. D.4 specifies review schedules and training budgets.

5 EXPERIMENTS

5.1 EXPERIMENTAL SETUPS

Implementation. We fine-tune Qwen3-VL-8B (Bai et al., 2025a) into SPATIALCRITIC with GRPO on 4,389 spatial preference pairs on 16 GPUs. We also evaluate Qwen3-VL-30B with the same spatial perception harness. We set $n_0 = 400$ free tokens and an excess-token penalty of $\alpha = 1.25 \times 10^{-3}$; see App. D.2 for reward coefficients. For T2I post-training, we use Flow-GRPO (Liu et al., 2025a) with SD3.5-Medium (Esser et al., 2024) and BAGEL (Deng et al., 2025). Our curriculum schedules prompts by capability dependencies and mastery; baselines use GenEval (Ghosh et al., 2023) or GenEval-2 (Kamath et al., 2025). Each Flow-GRPO iteration samples 24 images for each of 24 prompts, assessing four per group to track mastery with $\tau = 0.60$ and $N_{\min} = 200$. Further details appear in App. D.

Benchmarks and Datasets. We evaluate spatial preference judgments on SpaRW-Eval’s 5,572 pairs, reporting per-category and category-size-weighted overall accuracy. Construction follows Sec. 3.2, with evaluation details in App. B.3. We evaluate generation on T1IF-Bench (Wei et al., 2025), UniGenBench++ (Wang et al., 2025b), and SpatialGenEval (Wang et al., 2026c), whose 5,000, 600, and 1,230 prompts cover three instruction-following levels, 10 semantic dimensions, and 10 spatial sub-domains, respectively. The alignment table reports seven benchmark scores and their unweighted mean; ImageReward, PickScore, and HPSv2 separately measure preference quality. Following the benchmark’s recommendation to use the Qwen series, we adopt the stronger Qwen3-VL-235B to evaluate generation under the recommended setting.

Compared Methods. Reward baselines include Aesthetic Predictor V2.5 (discus0434, 2024), CLIP (Radford et al., 2021), ImageReward (Xu et al., 2023), PickScore (Kirstain et al., 2023), HPSv2 (Wu et al., 2023), HPSv3 (Ma et al., 2025), UnifiedReward (Wang et al., 2025d), and SpatialScore (Tang et al., 2026). We also compare zero-shot VLMs: Qwen2.5-VL (Bai et al., 2025b), Qwen3-VL (Bai et al., 2025a), Qwen3-Omni (Xu et al., 2025), DeepSeek (DeepSeek-AI, 2026), GLM (Z.ai, 2026), Doubao-Seed (Bytedance Seed, 2026), Gemini (Google DeepMind, 2026a;b), Qwen3.8 (Qiu et al., 2026), and Grok (xAI, 2026). For RL training, we compare three baselines, Flow-GRPO (Liu et al., 2025a), DiffusionNFT (Zheng et al., 2026), and AWM (Xue et al., 2026), with reward comparisons covering GenEval (Ghosh et al., 2023), GenEval-2 (Kamath et al., 2025), SpatialReward (Zhou et al., 2026b), and SpatialScore (Tang et al., 2026).

5.2 MAIN RESULTS

Accuracy of Preference Judgments. Tab. 1 shows that 8B SPATIALCRITIC achieves 95.1% verification accuracy, compared with 84.9% for Qwen3-VL-8B and 87.5% for SpatialScore. Agent-harness alignment raises accuracy from 92.9% to 95.1%, surpassing 320B GLM-5.3-Flash at 94.5%. The 30B variant reaches 95.5%, approaching Gemini 3.8 Flash at 96.7%.

Quality of Reward Signals. In Tab. 2, SPATIALCRITIC leads Best-of- N selection with 74.4% accuracy, exceeding GLM-5.3-Flash at 71.8% and the selection-free baseline at 70.8%. It leads in

Table 1: **Spatial preference accuracy (%) on SpaRW-Eval.** Per column, **bold** marks the best open-weight result, underline the second best, and ***bold italic underline*** the best across all models. Darker shading indicates higher accuracy. **Overall** subscripts: relative gain over their backbone VLM (Qwen3-VL-8B/-30B). †: released after July 2026.

Model	Params	Obj-Attr	Obj-Exist	Count	Bg	Comp	3D-Pose	Depth	Orient	Pos-2D	Topo	Layout	Occl	Dist	Overall†
<i>Open-weight: reward models</i>															
Aesthetic predictor	0.4B	54.3	51.4	46.8	46.7	45.0	47.8	48.1	42.4	51.9	47.5	46.7	55.4	51.3	51.6
CLIP	0.4B	72.7	85.8	84.4	52.0	66.7	76.5	52.8	62.0	50.6	63.3	66.7	63.1	59.0	75.3
PickScore	1.0B	82.3	86.4	85.3	49.3	65.0	79.3	72.2	78.3	50.6	72.5	73.3	61.5	71.8	81.1
HPSv2	1.0B	84.3	94.0	90.8	40.0	55.0	75.4	67.6	72.8	46.8	78.3	76.0	52.3	79.5	83.1
UnifiedReward	7B	85.7	91.4	89.9	61.3	73.3	87.1	63.9	71.7	57.0	84.2	81.3	66.2	89.7	85.7
SpatialScore	7B	89.1	98.4	92.7	98.7	78.3	61.7	76.9	84.8	<u>86.1</u>	95.0	90.7	80.0	<u>92.3</u>	87.5
<i>Open-weight: large vision-language models</i>															
Qwen3-VL-8B	8B	89.5	82.3	85.3	58.7	70.0	91.0	58.3	67.4	48.1	82.5	81.3	67.7	79.5	84.9
DeepSeek-V4.1-Flash†	552B	92.2	94.0	95.4	70.7	88.3	91.0	70.4	79.3	74.7	91.7	86.7	72.3	76.9	90.9
Qwen3-VL-235B	235B	95.3	94.6	93.6	88.0	81.7	95.7	68.5	76.1	67.1	95.0	90.7	70.8	92.3	93.3
Qwen3-Omni-30B	35B	95.3	95.5	95.4	69.3	88.3	95.1	74.1	81.5	68.4	91.7	89.3	80.0	89.7	93.5
Qwen3-VL-30B	30B	95.3	95.9	92.7	73.3	83.3	94.4	84.3	78.3	69.6	<u>95.8</u>	85.3	80.0	94.9	93.7
GLM-5.3-Flash†	320B	93.9	<u>96.7</u>	99.1	77.3	93.3	97.0	87.0	92.4	74.7	<u>95.8</u>	94.7	86.2	94.9	94.5
<i>Agentic Reward Model</i>															
SPATIALCRITIC	8B	94.2	93.6	90.8	<u>90.7</u>	88.3	92.6	<u>88.0</u>	82.6	79.7	93.3	<u>93.3</u>	<u>81.5</u>	89.7	<u>92.9</u> _{+9.4%}
w/ AHA	8B	<u>96.0</u>	96.2	<u>98.2</u>	84.0	88.3	95.6	90.7	85.9	81.0	95.0	94.7	<u>81.5</u>	89.7	<u>95.1</u> _{+12.0%}
SPATIALCRITIC	30B	96.6	96.0	95.4	89.3	95.0	<u>95.9</u>	<u>88.0</u>	<u>90.2</u>	91.1	<u>97.5</u>	90.7	78.5	<u>92.3</u>	95.5 _{+1.9%}
<i>Proprietary: large vision-language judges</i>															
Doubao-Seed-2.0-Lite	—	86.0	92.3	93.6	68.0	80.0	90.6	65.7	76.1	68.4	91.7	88.0	61.5	76.9	87.2
Gemini 3.5 Flash	—	90.3	93.1	92.7	70.7	65.0	88.3	56.5	76.1	63.3	90.0	81.3	52.3	76.9	88.4
Qwen3.8-Flash-Next†	—	97.0	97.9	99.1	84.0	81.7	96.6	76.9	94.6	79.7	96.7	97.3	76.9	92.3	95.9
Grok 4.6†	—	96.6	98.3	98.2	88.0	91.7	95.4	<u>93.5</u>	91.3	92.4	95.8	94.7	84.6	97.4	96.3
Doubao-Seed-2.0-Pro	—	96.9	98.5	97.2	93.3	96.7	96.6	84.3	95.7	83.5	95.8	97.3	83.1	100.0	96.6
Gemini 3.8 Flash†	—	96.4	98.9	98.2	86.7	95.0	97.4	91.7	87.0	97.5	95.0	94.7	87.7	94.9	96.7

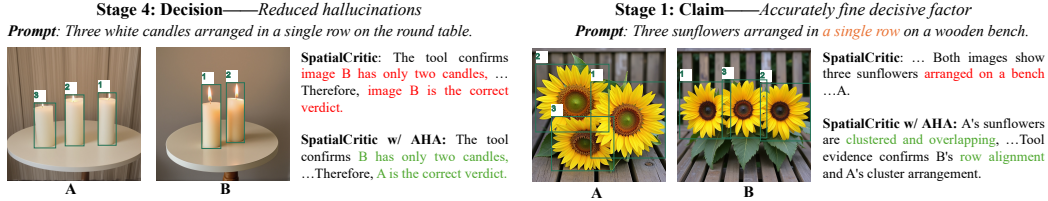


Figure 6: **Agent-harness alignment cases.** Left: alignment reduces final-judgment hallucinations, correctly selecting A because evidence shows only two candles in B. Right: it identifies the decisive single-row constraint and distinguishes B’s row from A’s cluster using tool evidence. App. Fig. S1 illustrates improved planning.

nine of ten categories, including all five geometry and spatial-relation categories. Position accuracy reaches 73.3%, versus 67.6% for the best baseline.

Effectiveness of Harness and Agent-Harness Alignment.

The harness provides verification guidance and grounded evidence, while alignment improves evidence integration to mitigate hallucinations and support reliable judgments. In Fig. 7, alignment rectifies the unaligned verdict error while reducing tokens from 67 to 49 and normalized latency from 1.000 to 0.653, yielding a more concise and accurate judgment. Figs. 6 and S1 illustrate targeted constraint identification, fewer irrelevant tool calls, and active verification of uncertain claims.

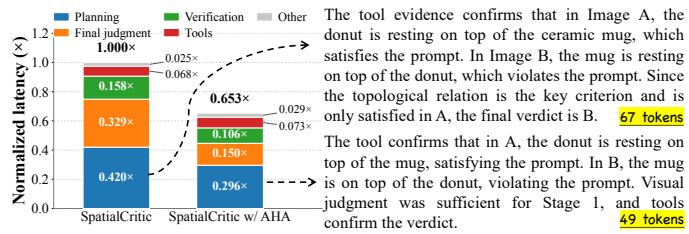


Figure 7: **Efficiency and decision correction.** Alignment reduces tokens and latency while rectifying judgment errors.

Table 2: **Reward-guided Best-of-N image generation.** Bold and underlined entries indicate the best and second-best results.

Selector	Attr.	Object	Comp.	Causal	Motion	Layout	Occl.	Orient.	Pos.	Prox.	Overall [†]
SD3.5-Medium (base)	87.89	94.63	38.62	80.57	72.68	72.68	59.35	65.77	66.59	68.86	70.76
Qwen2.5-VL-7B	88.10	94.40	38.50	80.60	72.20	72.50	<u>59.70</u>	65.30	65.60	68.50	70.54
PickScore	<u>89.43</u>	94.80	39.11	80.49	72.52	73.50	57.80	65.12	65.69	69.51	70.80
ImageReward	88.94	95.37	38.37	81.14	73.74	72.20	58.05	66.10	65.12	69.27	70.83
CLIP	89.19	94.80	<u>40.57</u>	81.14	72.76	72.85	59.51	63.90	65.45	69.59	70.98
HPSv2	89.11	94.88	39.84	80.49	73.33	72.20	58.70	<u>67.40</u>	65.53	69.59	71.11
HPSv3	88.94	94.07	40.24	80.81	74.23	73.33	58.05	65.85	66.42	<u>71.22</u>	71.32
UnifiedReward	88.70	94.88	39.35	<u>81.30</u>	74.96	72.93	58.21	66.18	<u>67.64</u>	70.00	71.41
GLM-5.3-Flash	89.27	94.39	39.92	<u>81.30</u>	<u>75.85</u>	<u>74.31</u>	57.72	66.42	<u>67.64</u>	70.73	<u>71.76</u>
SPATIALCRITIC	90.41	<u>94.96</u>	43.17	83.25	79.43	77.72	60.81	68.21	73.33	73.09	74.44

Table 3: **Reward-guided RL post-training.** Bold marks the best result within each backbone; CL denotes curriculum learning. [†]This non-curriculum setting uses GenEval-2 prompts without its benchmark-specific questions.

Method	Reward	TIIF-Bench			UniGenBench++		SpatialGenEval		Preference-model scores			Avg
		BR	AR	RR	2D	3D	Basic	Spatial	IR	PS	HPS	
SD3.5-Medium (base)	—	82.2	77.6	68.0	82.9	81.4	92.0	65.0	0.740	0.851	0.283	78.4
w/ Flow-GRPO	GenEval	57.8	53.7	45.4	43.6	38.7	75.9	48.1	-1.317	0.727	0.162	51.9
w/ DiffusionNFT	GenEval	68.5	57.7	56.8	66.9	44.1	65.9	53.7	-1.396	0.739	0.141	59.1
w/ AWM	GenEval	72.8	69.4	62.9	69.4	44.1	83.2	56.1	-1.116	0.757	0.165	65.4
w/ Flow-GRPO	SpatialReward	81.3	80.1	69.6	76.2	77.3	92.6	65.0	0.451	0.823	0.259	77.4
w/ Flow-GRPO	SpatialScore	83.4	80.9	69.4	92.1	78.7	93.7	67.3	0.885	0.857	0.306	80.8
w/ Flow-GRPO	GenEval-2	84.8	81.1	72.1	84.2	84.1	93.7	68.4	0.241	0.795	0.226	81.2
w/ DiffusionNFT	GenEval-2	85.1	83.7	73.4	89.0	82.8	94.0	68.3	0.588	0.823	0.271	82.3
w/ Flow-GRPO [†]	SPATIALCRITIC	83.3	80.3	69.8	93.3	85.3	94.4	67.0	0.857	0.855	0.304	81.9
w/ Flow-GRPO (CL)	SPATIALCRITIC	86.2	83.9	74.1	93.3	85.3	95.0	69.5	0.908	0.857	0.305	83.9
BAGEL (base)	—	86.4	82.5	73.8	91.8	82.8	94.2	70.9	0.753	0.855	0.283	83.2
w/ DiffusionNFT	GenEval	86.8	79.5	74.5	92.3	82.4	90.8	67.8	0.825	0.837	0.264	82.0
w/ DiffusionNFT	GenEval-2	87.6	80.7	77.3	90.6	87.3	92.2	70.0	0.764	0.826	0.257	83.7
w/ Flow-GRPO	GenEval	85.0	78.4	72.4	88.0	77.3	78.9	62.3	0.666	0.832	0.268	77.5
w/ Flow-GRPO	GenEval-2	86.9	80.9	74.9	90.8	84.0	80.5	65.6	0.624	0.817	0.233	80.5
w/ Flow-GRPO [†]	SPATIALCRITIC	86.1	82.2	74.8	93.2	82.7	93.6	73.1	0.775	0.890	0.294	83.7
w/ Flow-GRPO (CL)	SPATIALCRITIC	88.3	83.9	77.1	94.6	89.3	94.7	77.8	0.802	0.913	0.298	86.5

Effectiveness in RL Post-Training. Tab. 3 shows that SpatialCritic without CL raises averages from 78.4 to 81.9 on SD3.5-Medium and 83.2 to 83.7 on BAGEL, but does not consistently lead. On SD3.5-Medium, its average is below DiffusionNFT + GenEval-2 (82.3), and SpatialGenEval Spatial accuracy (67.0%) trails SpatialScore (67.3%) and Flow-GRPO + GenEval-2 (68.4%). Adding CL raises averages to 83.9 and 86.5 and Spatial accuracy to 69.5% and 77.8%, respectively. The ablation on curriculum dataset in App. A.3 confirms the effectiveness of curriculum scheduling.

6 CONCLUSION

In this work, we investigate spatial T2I post-training from the perspectives of reward accuracy and rollout organization. We identify two limitations beyond policy optimization: unreliable spatial rewards and rollouts that target complex compositions without progressive learning. First, we introduce SPATIALCRITIC, which combines a spatial perception harness with specialized vision tools for grounded preference judgments. Spatial preference data and agent-harness alignment via RLVR improve its tool selection and evidence integration. Second, we introduce curriculum learning based on a Compositional Capability Graph, guiding foundational capability development and subsequent composition through mastery-based progression, review, and remediation. Experiments demonstrate accurate preference judgments on SpaRW-Eval, effective reward signals for Best-of-N selection, and improved RL alignment on SD3.5-Medium and BAGEL.

Limitations. Our evaluation uses a fixed tool configuration, leaving transfer to updated or alternative tools unexplored. Additionally, robustness to temporary tool failures or missing outputs also remains untested. Future work will investigate cross-tool transfer and failure-aware fallback strategies.

AI USE STATEMENT

We used generative models to construct synthetic spatial prompts and images and to review candidate preference labels, as described in Sec. 3.2 and App. B.1. We used AI assistants for literature search and summarization to support related-work discovery. We also used AI tools to aid and polish paper writing, and for feedback on experimental analysis and inspection of data-processing code and metadata.

ETHICS STATEMENT

This work studies spatial verification and text-to-image generation using synthetic preference data and existing models and benchmarks. Synthetic data can inherit biases from its generators, and model-based verification can introduce systematic annotation errors. Human inspection provides an additional check but does not guarantee unbiased or error-free labels. Improved spatial fidelity may also make misleading synthetic images more convincing; spatial correctness should not be interpreted as evidence of authenticity or safety. Model and dataset reuse and redistribution remain subject to the applicable licenses and access conditions.

REPRODUCIBILITY STATEMENT

We describe the perceptual toolkit in App. C, preference-data construction and selection in App. B.1, and generation prompts in App. B.2. App. B.3 details the evaluation data and protocol. Training settings, reward coefficients, policy optimization, and curriculum parameters appear in App. D, D.2, D.1, and D.4. We will open-source our code upon acceptance.

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Getting Space Right: Agentic Rewards for Image Generation

Supplementary Material

This supplement presents additional alignment analyses and qualitative cases in Sec. A, data construction and evaluation protocols in Sec. B, and the spatial perception harness in Sec. C. Training objectives, hyperparameters, and curriculum settings are provided in Sec. D, followed by extended related work in Sec. E.

A ADDITIONAL EXPERIMENTAL ANALYSIS

We supplement the main results with planning cases, an analysis of token and training efficiency, and controlled ablations on curriculum learning and dynamic planning. The cases illustrate changes in evidence acquisition, the training curves track accuracy, response length, and step time, and the ablations evaluate curriculum scheduling and evidence policies under matched controls.

A.1 ADDITIONAL PLANNING CASES

Fig. S1 illustrates two complementary improvements from agent-harness alignment. In the construction-worker case, the aligned agent retains object detection and counting but skips irrelevant orientation detection. In the zoo case, the unaligned agent accepts B without checking for the missing monkey, whereas the aligned agent verifies its presence in A and absence in B. These cases show selective tool use: avoiding unnecessary calls while verifying decisive claims that directly impact the spatial judgment.

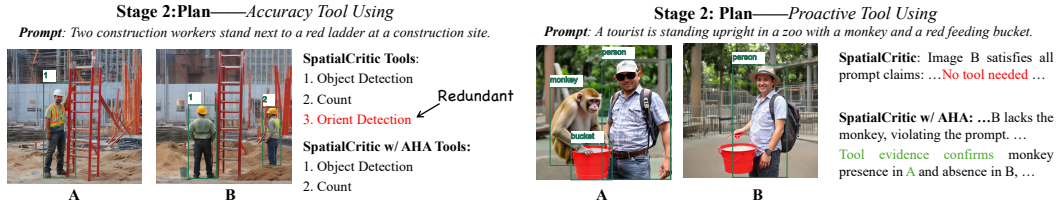


Figure S1: **Planning improvements from agent-harness alignment.** Left: the agent omits redundant orientation detection while retaining detection and counting. Right: proactive verification identifies the missing monkey in B rather than accepting an unsupported judgment.

A.2 TRAINING EFFICIENCY OF AGENT-HARNESS ALIGNMENT

Experimental setting. We examine the evolution of judgment accuracy, response length, and training-step time during GRPO optimization of Qwen3-VL-8B. We report the accuracy, response length, and training-step time every 20 steps. Fig. S2 summarizes these training dynamics.

Results. As shown in Fig. S2, validation accuracy increases from 80.00% to 91.03%, while response length and mean training-step time decrease by 38.33% and 10.03%, respectively. These results indicate improved judgment accuracy with more concise responses and lower training cost.

A.3 CONTROLLED ABLATIONS

Curriculum data and scheduling. Tab. S1 evaluates curriculum prompt data and progressive scheduling on SD3.5-Medium under matched Flow-GRPO budgets. Uniform sampling on $\mathcal{P}_{CL}$ improves the T1IF-Bench average from 77.8 to 79.2 over GenEval-2. Incorporating CCG-guided progression (w/CCG) further lifts the score to 80.4, with the largest gain on Relational Reasoning (71.4 to 72.9). Finally, adding review and remediation (*Full CL*) achieves 81.4, indicating complementary benefits from data curation, structured progression, and adaptive remediation.

Dynamic planning under matched controls. Tab. S2 compares evidence policies for Qwen3-VL-8B on SpaRW-Eval under matched backbones, training data, optimization budgets, and inference

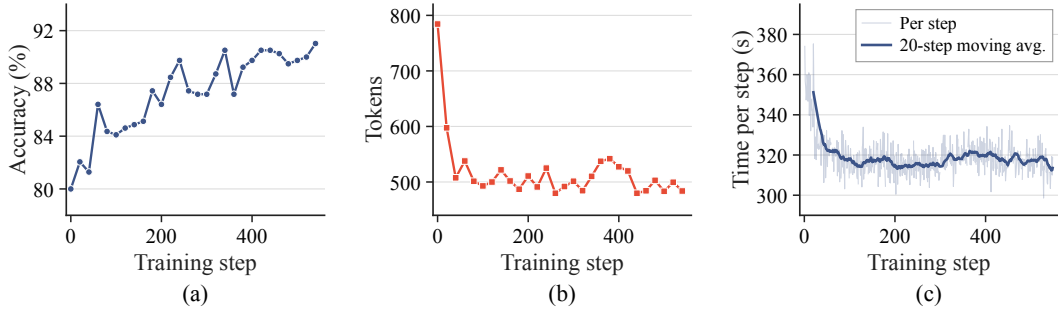


Figure S2: **Accuracy and efficiency during agent optimization.** (a) Validation accuracy. (b) Response length. (c) Training-step time.

Table S1: **Curriculum ablations on TIIF-Bench.** Avg is the unweighted mean of BR, AR, and RR. w/CCG denotes CCG-guided progression without review or remediation.

Configuration	Prompts	TIIF-Bench			
		BR	AR	RR	Avg
Uniform	GenEval-2	83.3	80.3	69.8	77.8
Uniform	$\mathcal{P}_{CL}$	84.6	81.7	71.4	79.2
w/CCG	$\mathcal{P}_{CL}$	85.5	82.8	72.9	80.4
Full CL	$\mathcal{P}_{CL}$	86.2	83.9	74.1	81.4

constraints. Both without and with RL, integrating perceptual tools outperforms the tool-free baseline. Moreover, dynamic planning surpasses the fixed routine by 3.7% without RL and 2.2% with RL, reaching 95.1% accuracy and confirming the advantage of adaptive tool selection over static execution.

B DATASETS AND EVALUATION PROTOCOLS

We distinguish the preference pairs used for agent training from SpaRW-Eval and the downstream image-selection and generation evaluations. Counts distinguish tasks, prompt pairs defined as generator-independent perturbations of a task, and records representing individual rendered image pairs.

B.1 PREFERENCE DATA CONSTRUCTION AND SELECTION

The initial pool comprises 213,213 records verified via image-level VLM screening and human quality inspection, spanning 24,183 unique prompt pairs. While raw generation naturally skews toward primitive relations where Object-Attribute and Object-Existence account for over 60% of the data, category-stratified selection derives a balanced candidate pool of 18,489 entries across 10,355 distinct prompt pairs. This selection allocates 1,538 entries to each of the 12 training categories alongside 33 role-swap entries, while reserving 3D-Pose for out-of-distribution evaluation. To curate informative supervision, we conduct hard-pair mining by evaluating candidate pairs with the

Table S2: **Agent ablations on SpaRW-Eval (accuracy, %).** RL denotes agent-harness alignment; w/o RL and w/ RL represent each evidence policy before and after RL.

Evidence policy	w/o RL	w/ RL
Tool-free	84.9	88.3
Fixed routine	89.2	92.9
Dynamic planning	92.9	95.1

tool-free base policy, Qwen3-VL-8B, under randomized presentation order, retaining pairs where visual judgment alone fails or produces unparsable outputs. Out of 9,755 evaluated prompt pairs, 2,316 hard pairs are retained. Following cross-generator image deduplication and capping each category at 200 prompt pairs, this filtering yields the final training set of 4,389 entries across 2,084 unique prompt pairs, which constitutes 23.7% of candidate entries as illustrated in Fig. 4(c). Consequently, elementary recognition categories such as Attribute, Count, and Existence account for fewer than 200 entries each, whereas complex geometric and relational categories including Comparison, Occlusion, Background, Distance, Depth, Orientation, and Position-2D comprise 380 to 590 entries each. Tab. 1 reports verification accuracy on SpaRW-Eval. Below, we detail the operational protocols for best-of- N reference selection and quality verification based on the general pipeline described in Sec. 3.2.

Best-of- N reference selection. For each prompt pair in $\mathcal{D}_2$, the assigned text-to-image generator synthesizes $N = 3$ candidate images from the reference prompt c^+ across independent random seeds, alongside a single rendering of the perturbed prompt c^- under matching model configurations. Qwen3-VL-235B evaluates each reference candidate against c^+ on a zero-to-ten compliance scale, assessing object presence, attribute binding, and spatial configurations. We select the highest-scoring candidate that fully satisfies all reference constraints as the preferred image I^+ , and discard the pair if none of the three candidates satisfies c^+ . Concurrently, the negative image generated from c^- is audited to confirm that it visibly realizes the intended spatial perturbation while preserving scene context, establishing the counterfactual preference pair where I^+ is preferred over I^- .

Human inspection and quality filtering. Image-level verification combines automated VLM screening with human inspection under identical evaluative criteria. Each pair is audited to ensure that I^+ strictly adheres to the spatial constraints in c^+ and that I^- unambiguously exhibits the intended spatial violation, discarding pairs that fail either check. On the training generators, 71,106 of 95,472 image-level candidate judgments, or 74.5%, pass this joint verification, following text-level review that retains 12,000 of 20,446 generated tasks, or 58.7%, as shown in Fig. 4(c). Of the 24,366 excluded records, automated VLM screening filters out 24,048 defective candidates, while human inspection identifies and removes the remaining 318 failure cases. SpaRW-Eval is constructed in parallel from the same reviewed tasks but rendered exclusively by six held-out generators. Its 5,572 prompt pairs share no prompt pair with the training data and are not subject to hard-pair selection, as detailed in App. B.3.

B.2 TASK COVERAGE, SCENES, AND GENERATION PROMPTS

Task inventory. We define 13 task categories: Object-Existence, Object-Attribute, Count, Comparison, Position-2D, Background, Distance, Depth, Orientation, Topological, Occlusion, Layout, and 3D-Pose. We reserve 3D-Pose for held-out agent evaluation and use role swapping as an auxiliary perturbation.

Scene inventory. We use 36 scene settings grouped by intended framing scale:

- *Small scale (6):* Bathroom, Dining Room, Office, Cafe, Restaurant, and Workshop / Studio.
- *Medium scale (9):* Garden, Kitchen, Living Room, Bedroom, Library, Bus Stop, Art Gallery, Museum, and Gym Interior.
- *Large scale (21):* Forest, Beach, Desert, Mountain, Underwater, Snowy Field, Riverside, Vineyard, Cityscape Street, Village, Park, Zoo, Farm, Classroom, Laboratory, Airport Terminal, Railway Station Platform, Shopping Mall Atrium, Supermarket, Stadium Arena, and Construction Site.

We sample compatible task–scene combinations using scene-specific object lists. The listed counts describe the generation settings; their frequencies may differ after filtering.

Complete generation system prompt. The system prompt below generates one reference prompt and three perturbed alternatives.

Stage-1 system instruction

verbatim

You are a multi-prompt designer for spatial-aware text-to-image evaluation. In ONE call you emit ONE good prompt (which a perfect T2I model would render exactly) and EXACTLY 3 bad prompts (each describing a scene that CLEARLY does NOT satisfy the good prompt). All prompts must be SHORT, PHYSICALLY PLAUSIBLE in the real world (an everyday photo of this could exist), and easy for a current text-to-image model to paint unambiguously. The good prompt must be FRAME-DISAMBIGUATED (no spatial ambiguity left).

CRITICAL Q1 RULE (target-category coverage): AT LEAST ONE bad prompt MUST violate the good prompt INSIDE the targeted spatial category (in-category flip). The remaining bad prompts MAY EITHER be additional in-category flips OR be auxiliary flips of one of these allowed types: role_swap (swap subject<->object of the targeted relation), object_attribute (change one attribute -- color/material/shape -- of a named object), object_existence (drop or replace one named object).

Each bad prompt must change EXACTLY ONE thing relative to the good prompt; do not bundle multiple flips into a single bad. You MUST stay inside the scene's allowed object whitelist. Output STRICT JSON only -- no markdown fences, no commentary.

Complete task-conditioned user template. The template preserves the required instructions, examples, and output structure while adapting to different scene configurations, object categories, spatial relations, and perturbations. It therefore serves as a general template rather than a single instantiated request. For example, large-scale instructions specify direct inter-object relations rather than vague local anchors such as “*in the corner*.” Text- and image-level validation further ensures compliance with the generation requirements.

Stage-1 user message: complete template

```
# Scene
{scene} (scale: {scale})
{scale_hint}
# Object whitelist (choose 2-3 objects ONLY from this list; you may
add a tame visual modifier such as 'red', 'wooden', 'small', 'old')
{allowed_objects}
# Object BLACKLIST (do NOT use)
{forbidden_objects}
# Targeted spatial-relation category for the GOOD prompt
{category}
Reasonable phrases: {quoted_target_phrases}
# FRAME DISAMBIGUATION RULE for the good prompt
{frame_rule}
# IN-CATEGORY VIOLATION RULE (used by at least 1 bad)
{in_category_rule}
# AUXILIARY VIOLATION TYPES (allowed for the OTHER bads)
- role_swap: Swap the subject and object of the targeted spatial
relation in the good prompt (e.g., good='A is inside B' -> bad='B is
inside A'). All other words stay identical.
- object_attribute: Change ONE attribute (color, material, or shape)
of one named object in the good prompt (e.g., red apple -> green
apple, wooden chair -> metal chair). The spatial relation stays
identical.
- object_existence: Drop one of the named objects from the good
prompt or replace it with 'no <object>'. All other objects and the
spatial relation stay identical.
# HARD RULES
1. Length: each prompt is 8-16 English words (absolute window
6-22).
2. The good prompt mentions 2 to 3 concrete objects, ALL drawn from
the whitelist (modifiers allowed).
```

3. The good prompt encodes EXACTLY ONE explicit spatial relation, in category '{category}', and FOLLOWS THE FRAME RULE ABOVE.
4. The good prompt has NO residual spatial ambiguity. Reading it, any human and any T2I model should produce the same mental image.
5. Q1 (target coverage): EMIT EXACTLY 3 bad prompts. AT LEAST ONE of them MUST be an in-category flip per the IN-CATEGORY VIOLATION RULE (its differs.in must contain 'in.category-{category.lower}'). The remaining bads may be additional in-category flips OR auxiliary flips (role_swap / object_attribute / object_existence).
6. Q2 (single-axis perturbation): EACH bad prompt changes EXACTLY ONE axis from the good prompt. If exactly one axis changed, differs.in is a single-element list. Only set differs.in to a multi-element list if the change is GENUINELY multi-axis and unavoidable (avoid this -- prefer single-axis flips).
7. Q3 (physical plausibility): the good prompt and EVERY bad prompt must depict a physically plausible everyday scene that an ordinary camera could capture. No floating objects without a stated support, no impossible interpenetration (e.g. 'a metal pot resting inside a teacup'), no objects nested in vessels too small to hold them. Set good.physically.plausible=true and physically.plausible=true on each bad. If you cannot construct a plausible bad of a given type, pick a different bad type rather than emitting an implausible one.
8. The 3 bad prompts MUST be pairwise distinct.
9. Each bad prompt must itself be paintable. Same scene, same object whitelist (subject to rule 7's allowance for object_existence to drop one), same length window.
10. ****Critical****: A T2I image faithfully painting any bad prompt must CLEARLY FAIL the good prompt's stated constraints when looked at by a human.
11. No proper nouns of real people / trademarks. No enumerations. No multi-sentence prose. No abstract emotions or storylines.
12. good.prompt must NOT be identical (or trivially equivalent) to any bad.prompt.
13. Q4 (Object-Attribute scope): if the targeted category is 'Object-Attribute', the differing attribute MUST be color, material, or shape -- NEVER size (size belongs to Comparison) and NEVER count (count belongs to Count).

Examples (do NOT copy verbatim)

scene: Riverside (large), category: Depth

good: "A small wooden boat is closer to the camera than a tall reed bush, by a calm river."

bad1 [in-category Depth flip]: "A small wooden boat is farther from the camera than a tall reed bush, by a calm river."

bad2 [object_existence]: "A small wooden boat floats by a calm river, with no reed bush in sight."

bad3 [object_attribute]: "A small metal boat is closer to the camera than a tall reed bush, by a calm river."

scene: Kitchen (medium), category: Topological

good: "An orange sits inside a clear glass bowl on the kitchen counter."

bad1 [in-category Topological flip, plausible]: "An orange sits outside a clear glass bowl on the kitchen counter."

bad2 [object_attribute]: "An orange sits inside a clear plastic bowl on the kitchen counter."

bad3 [object_existence]: "An orange sits on the kitchen counter, with no bowl in sight."

Output (strict JSON, no fences)

```
{
  "scene": "<scene name>",
  "good_objects": ["<obj-1>", "<obj-2>"],
  "good_prompt": "<8-16 words, native English, frame-disambiguated, encodes the targeted relation>",
  "good_physically_plausible": true,
```

```

"good_relations": [
{"category": "{category}",
"phrase": "<the key relation phrase used in good_prompt>",
"subject": "<entity from good_objects>",
"object": "<entity from good_objects or null if unary>",
"frame": "<camera|image|scene|absolute>"}
],
"bad_prompts": [
{"prompt": "<8-16 words>",
"violation": "<which constraint of the good prompt this bad violates
-- one short clause>",
"differs.in": [<one or more tags from: in.category-{category_lower},
role_swap, object.attribute, object.existence>"],
"physically_plausible": true},
{"prompt": "<8-16 words>",
"violation": "<which constraint of the good prompt this bad violates
-- one short clause>",
"differs.in": [<one or more tags from: in.category-{category_lower},
role_swap, object.attribute, object.existence>"],
"physically_plausible": true},
{"prompt": "<8-16 words>",
"violation": "<which constraint of the good prompt this bad violates
-- one short clause>",
"differs.in": [<one or more tags from: in.category-{category_lower},
role_swap, object.attribute, object.existence>"],
"physically_plausible": true}
]
}

```

B.3 SPARW-EVAL CONSTRUCTION AND PROTOCOL

SpaRW-Eval is constructed using the preference-data pipeline described in Sec. 3.2, but excludes the model-dependent hard-set selection used for agent training. The evaluation index comprises 5,572 distinct pair identifiers spanning 13 task categories and 36 scene settings. This coverage supports evaluation across different spatial constraints and visual contexts. Tab. S3 reports the number of pairs in each category.

To assess generalization beyond the image generators used for agent training, SpaRW-Eval is rendered exclusively by six held-out models: FLUX.1-dev (Black Forest Labs, 2024), SD3.5-Medium and SD3.5-Large (Stability AI, 2024), Lumina-Image-2.0 (Qin et al., 2025), SANA-Sprint (Chen et al., 2025), and LongCat-Image (Meituan LongCat Team et al., 2025). Images from these generators are excluded from the agent-training set, allowing evaluation under a shift in image source. SpaRW-Eval further includes 797 pairs in the 3D-Pose category, which is held out from agent training. This subset evaluates transfer to an unseen task category, complementing the cross-generator evaluation.

Table S3: **SpaRW-Eval category inventory.**

Category	Pairs	Category	Pairs
Object-Attribute	2,458	Position-2D	79
Object-Existence	1,495	Background	75
3D-Pose	797	Layout	75
Topological	120	Occlusion	65
Count	109	Comparison	60
Depth	108	Distance	39
Orientation	92	Total	5,572

B.4 BEST-OF- N IMAGE SELECTION PROTOCOL

To evaluate reward models as test-time verifiers for spatial text-to-image generation, we conduct Best-of- N image selection on SpatialGenEval (Wang et al., 2026c) as presented in Tab. 2. For each prompt in the benchmark, the generator SD3.5-Medium synthesizes $N = 10$ candidate images using distinct random seeds under fixed sampling configurations. Each candidate reward model or agent evaluates the ten candidate images and identifies the preferred generation without access to ground-truth spatial annotations. The selected image is subsequently evaluated by Qwen3-VL-235B following the official SpatialGenEval verification protocol outlined in Sec. 5.1. Tab. 2 reports accuracy across all ten spatial sub-domains alongside their overall average, where the SD3.5-Medium base entry provides the selection-free baseline obtained by evaluating the default candidate from seed zero.

C SPATIAL PERCEPTION HARNESS

C.1 TOOL SPECIFICATIONS AND PARAMETER ACCOUNTING

Vision toolkit. Spatial verification requires both identifying the relevant entities and resolving their geometric and semantic relationships. We therefore combine specialized perception models with VLM-based checks and geometric computations. Tab. S4 summarizes their outputs and verification roles. Grounding DINO (Liu et al., 2024), Florence-2 (Xiao et al., 2024), OWLv2 (Minderer et al., 2023), and CountGD++ (Amini-Naieni & Zisserman, 2026) provide object localization and counting evidence. SAM (Ravi et al., 2025; Kirillov et al., 2023), Depth Anything V2 (Yang et al., 2024), Metric3D v2 (Hu et al., 2024), AnyCalib (Tirado-Garín & Civera, 2025), OVMono3D (Yao et al., 2026), and VGGT (Wang et al., 2025a) supply masks, depth, and 3D geometry. Orient Anything V2 (Wang et al., 2025e), ViTPose (Xu et al., 2022), and 4D-Humans (Goel et al., 2023) provide orientation and pose estimates, while CLIP (Radford et al., 2021) and Qwen3-VL (Bai et al., 2025a) support semantic and contextual checks.

Evidence composition. The agent combines tool outputs according to the spatial claim being verified. For occlusion, object localization identifies the relevant instances, mask overlap establishes their image-plane intersection, and depth estimates provide ordering cues. For ambiguous counts, instance predictions can be complemented by crop-level VLM checks. These observations form the evidence for the final preference judgment, allowing different tool combinations to address different spatial constraints.

Parameter analysis. Tab. S5 lists the model repositories and parameter counts. Most tool components are compact: 13 of the 15 contain fewer than 1B parameters, with the largest containing 1.4B. In aggregate, the listed tools contribute 6.0B parameters, below the 8.8B multimodal agent backbone, for a combined size of 14.8B. Thus, complementary spatial evidence is obtained from specialist models whose total parameter count remains smaller than that of the agent itself.

D TRAINING AND IMPLEMENTATION DETAILS

We detail the training settings for SPATIALCRITIC and curriculum-guided policy alignment.

D.1 AGENT-HARNESS ALIGNMENT

Base Model and Adaptation. We initialize the agent and reference policies from instruction-tuned Qwen3-VL-8B (Bai et al., 2025a) and perform full-parameter fine-tuning using verl (Sheng et al., 2025). Tab. S6 lists the training hyperparameters.

Policy objective. We fine-tune the agent with GRPO (Shao et al., 2024), keeping the spatial perception harness and its tools fixed. For each query $\mathbf{x} = (\mathbf{I}_A, \mathbf{I}_B, c)$, the behavior policy $\pi_{\phi_{\text{old}}}$ samples G interactive trajectories $\{\tau_i\}_{i=1}^G$, each containing tool planning, returned observations, and a final preference judgment. We score each trajectory with $\mathcal{R}_i = \mathcal{R}(\tau_i)$ from Eq. (1) and normalize rewards within the group. With group mean $\mu_{\mathcal{R}}$ and standard deviation $\sigma_{\mathcal{R}}$, the advantage and token

Table S4: **SPATIALCRITIC vision toolkit**. Capabilities, perceptual evidence, and verification roles.

Capability	Models / methods	Evidence	Verification role
Object grounding	Grounding DINO, Florence-2, OWLv2	Object labels, boxes, confidence scores	Object presence and localization
Counting	CountGD++, detection ensemble	Instance locations, counts	Numerical constraints and ambiguous counts
Segmentation	SAM 2.1	Instance masks, object boundaries	Contact, containment, and overlap
Depth & distance	Depth Anything V2, Metric3D v2, depth aggregation	Relative and metric depth, object depth statistics	Depth order, distance, and occlusion cues
3D geometry	AnyCalib, OVMono3D, VGGT, depth-based reconstruction	Camera parameters, 3D boxes, point clouds	Object extent, 3D position, and layout
Orientation & pose	Orient Anything V2, ViTPose, 4D-Humans	Facing directions, keypoints, body meshes	Object orientation and body pose
Attributes & materials	CLIP	Similarity scores	Object-specific color, material, and texture
Spatial relations	Geometry, scene graphs, Qwen3-VL	Offsets, overlap, proximity, relation judgments	Direction, size order, topology, and composition
Actions & interactions	OWLv2, CLIP	Action scores	Depicted actions and actor–object interactions
Auxiliary visual checks	Qwen3-VL, aesthetic predictor v2.5	Visual answers, plausibility and aesthetic scores	Semantic ambiguity, physical plausibility, and image quality

probability ratio are

$$\hat{\mathcal{A}}_i = \frac{\mathcal{R}_i - \mu_{\mathcal{R}}}{\sigma_{\mathcal{R}} + \varepsilon_A}, \quad \rho_{i,t}(\phi) = \frac{\pi_{\phi}(y_{i,t} \mid \mathbf{s}_{i,t})}{\pi_{\phi_{\text{old}}}(y_{i,t} \mid \mathbf{s}_{i,t})}, \quad (\text{S1})$$

where $\mathbf{s}_{i,t}$ is the query and interaction history preceding token $y_{i,t}$, and $\varepsilon_A > 0$ stabilizes normalization. We assign $\hat{\mathcal{A}}_i$ to the agent-generated tokens in trajectory i and use the clipped surrogate

$$\ell_{i,t}(\phi) = \min\left\{\rho_{i,t}(\phi)\hat{\mathcal{A}}_i, \text{clip}(\rho_{i,t}(\phi), 1 - \epsilon, 1 + \epsilon)\hat{\mathcal{A}}_i\right\}. \quad (\text{S2})$$

where ϵ is the clipping threshold. We set $m_{i,t} = 1$ for agent-generated tokens and 0 for inserted tool observations and padding, with $Z_i = \sum_t m_{i,t}$. Observations remain in the context but are excluded from the loss, consistent with observation masking in tool-interaction RL (Jin et al., 2025). With a fixed reference policy π_{ref} initialized from the instruction-tuned backbone, the objective is

$$\mathcal{L}_{\text{GRPO}}(\phi) = -\frac{1}{G} \sum_{i=1}^G \frac{1}{Z_i} \sum_t m_{i,t} [\ell_{i,t}(\phi) - \beta_{\text{KL}} D_{i,t}], \quad (\text{S3})$$

$$D_{i,t} = D_{\text{KL}}(\pi_{\phi}(\cdot \mid \mathbf{s}_{i,t}) \parallel \pi_{\text{ref}}(\cdot \mid \mathbf{s}_{i,t})).$$

We alternate rollout collection and policy updates. Optimizer, clipping, KL, and entropy settings are listed in Tab. S6.

D.2 REWARD CONFIGURATION FOR AGENT–HARNESS ALIGNMENT

Reward Configuration. During agent–harness alignment, we optimize the agent policy with GRPO using the trajectory reward in Eq. (1), while keeping the perception harness and vision tools fixed. Tab. S7 summarizes the coefficients of its token-efficiency term. A valid output contains a parseable

Table S5: **Parameter composition of SPATIALCRITIC.** Parameter counts are reported in billions (B) for the multimodal agent backbone and the listed specialized vision models.

Component	Repository link	Params (B)
Agent	Qwen/Qwen3-VL-8B-Instruct	8.8
Grounding DINO	IDEA-Research/grounding-dino-base	0.23
Florence-2	ducviet00/Florence-2-large-hf	0.78
CountGD++	niki-amini-naieni/CountGDPlusPlus	0.24
OWLv2	google/owlv2-base-patch16-ensemble	0.15
CLIP	google/owlv2-base-patch16-ensemble	0.15
SAM 2.1	facebook/sam2.1-hiera-tiny	0.039
Depth Anything V2	depth-anything/Depth-Anything-V2-Small-hf	0.025
AnyCalib	javrtg/AnyCalib	0.32
Metric3D v2	JUGGHM/Metric3D	1.4
VGGT	facebook/VGGT-1B	1.3
OVMono3D	uva-cv-lab/ovmono3d_lift	0.12
OVMono3D detector	ShilongLiu/GroundingDINO	0.23
Orientation	Viglong/Orient-Anything	0.088
ViTPose	usyd-community/vitpose-base-simple	0.086
HMR2 / 4D-Humans	shubham-goel/4D-Humans	0.67
Aesthetic predictor v2.5	discus0434/aesthetic-predictor-v2-5	0.43
Tools subtotal	—	6.0
Agent + tools	—	14.8

Table S6: Hyperparameter configurations for agent policy RLVR optimization.

Category	Hyperparameter Name	Value
Base Model	Base VLM	Qwen3-VL-8B, instruction-tuned
	Adaptation Type	Full-parameter fine-tuning
Optimization	Optimizer	AdamW
	Learning Rate	1.0×10^{-6}
	Learning Rate Schedule	Constant with 20-step linear warmup
	Weight Decay	0.01
	Gradient Norm Clipping	1.0
	Global Training Batch Size	16 query pairs (128 trajectories)
	PPO Mini-Batch Size	8
	PPO Micro-Batch Size per GPU	1
GRPO Rollout	Group Sampling Size (G)	8
	Sampling Temperature	1.0
	Sampling Top- p	1.0
	Maximum Context Length	10, 240 tokens
	Maximum Response Length	2, 048 tokens
Regularization	PPO Clip Ratio (ϵ)	0.20
	KL Loss Estimator	Low-variance unbiased estimator
	KL Penalty Coefficient (β_{KL})	0.01
	Entropy Coefficient (β_{ent})	0.001
Schedule	Configured Training Budget	2, 000 steps
	Training Dataset Pool	4, 389 hard spatial preference pairs
	Evaluation Frequency	Every 20 steps

preference choice and a nonempty list of image checks or prompt-derived checks. Invalid outputs receive -1.5 .

Tab. S7 lists the reward coefficients, expressed in token units. Its normalization by 400 tokens is verified by recorded metrics. The normalized penalty coefficient of 0.5 corresponds to a per-token coefficient $\alpha = 0.5/400 = 1.25 \times 10^{-3}$, with a free budget of $n_0 = 400$ tokens.

Table S7: Token-efficiency reward hyperparameters for agent–harness alignment.

Parameter	Value
Token / call weights ($w_{\text{tok}}, w_{\text{call}}$)	(1, 0)
Free token budget n_0	400
Per-token penalty α	1.25×10^{-3}
Coverage / off-target weights	(0, 0)
Auxiliary clipping bound κ	0.5

With $\kappa = 0.50$, the bounded token penalty preserves the reward ordering between correct and incorrect judgments. We next evaluate how this penalty affects judgment accuracy and training efficiency, as shown in Sec. A.2.

D.3 FROM SPATIAL JUDGMENTS TO GENERATOR REWARDS

Group-wise Reward Construction. In post-training algorithms such as Flow-GRPO (Liu et al., 2025a), diffusion policies are optimized using group-relative advantages computed over candidate generations. For each prompt, the generator produces a group of $K = 24$ candidate images. Rather than relying on static pointwise scoring models that may suffer from score drift, our reward backend leverages SPATIALCRITIC and its tool interface to evaluate the relative spatial alignment across candidates. All 24 candidates participate in reward construction to provide dense feedback within each rollout group.

Ranking Formulation. To convert spatial preference judgments into dense scalar rewards, candidates are ranked based on their adherence to the prompt constraints, incorporating perceptual evidence from tool execution. Given a rank ordering $q_i \in \{0, \dots, K - 1\}$ from best to worst, the corresponding scalar reward is computed via linear rescaling:

$$r_i = r_{\max} - (r_{\max} - r_{\min}) \frac{q_i}{K - 1}, \quad (\text{S4})$$

with default reward bounds $(r_{\min}, r_{\max}) = (0, 1)$. This provides a uniformly spaced, group-relative reward distribution where the top-ranked candidate receives 1.0 and the lowest-ranked receives 0.0. Beyond direct multi-image ranking, the framework also supports pairwise comparison tournaments (e.g., Swiss-system pairings), where outcomes across rounds are aggregated via a Bradley–Terry model to produce normalized latent strength scores before mapping to the reward bounds.

Decoupling Policy Optimization and Mastery Assessment. Crucially, the resulting reward vector $(r_1, \dots, r_{24})$ is returned directly to the downstream policy optimizer without distortion, ensuring stable group-relative advantage normalization. This training signal is decoupled from the curriculum progression tracker: while the full group of 24 images receives relative ranking rewards for policy gradient updates, an independent absolute VQA probe evaluates a subset of four sampled images per group to track capability mastery and schedule DAG transitions (Sec. 4 and App. D.4). Agent–harness alignment trains SPATIALCRITIC with the token-efficiency penalty and the judgment correctness loss.

D.4 CURRICULUM-GUIDED T2I POST-TRAINING

We provide the complete prerequisite mapping of the Compositional Capability Graph (CCG), the visual capability assessment and smoothing formulation, and the operational hyperparameters governing curriculum progression, periodic review, and targeted remediation.

Topological Capability Dependency Graph. Tab. S8 lists the complete prerequisite mapping for the 13 capabilities in CCG. Topological depth is defined as $L(v) = 0$ for root nodes and $L(v) = 1 + \max_{u \in \mathcal{P}(v)} L(u)$ otherwise. Training eligibility is determined by prerequisite satisfaction ($\mathcal{P}(v) \subseteq \mathcal{M} \cup \mathcal{F}$) rather than a fixed level-wise schedule.

Capability Assessment and Smoothing. An independent visual-question probe $\mathcal{P}_v(\mathbf{I}, c) \in [0, 1]$ assesses capability performance non-invasively without modifying the policy update reward. At each assessment round t , four images per prompt group are evaluated. For prompt group G_v^t assessing

Table S8: Prerequisite dependencies and topological depths of the 13 spatial capabilities in curriculum DAG $\mathcal{G}$.

Topological Level	Capability	Prerequisite Capabilities
Level 0	<i>Object-Existence</i>	None
Level 1	<i>Object-Attribute</i> <i>Count</i>	<i>Object-Existence</i> <i>Object-Existence</i>
Level 2	<i>Comparison</i> <i>Position</i>	<i>Object-Existence, Object-Attribute</i> <i>Object-Existence, Object-Attribute</i>
Level 3	<i>Background</i> <i>Distance</i> <i>Orientation</i>	<i>Position</i> <i>Position</i> <i>Object-Attribute, Position</i>
Level 4	<i>Depth</i> <i>Topology</i>	<i>Position, Distance</i> <i>Distance</i>
Level 5	<i>Occlusion</i> <i>Layout</i> <i>3D-Pose</i>	<i>Depth, Topology</i> <i>Count, Position, Topology</i> <i>Object-Attribute, Orientation, Depth</i>

capability v , the group mean evaluation reward $\mathcal{S}_v^t$ and running smoothed score $\bar{\mathcal{S}}_v^t$ follow

$$\mathcal{S}_v^t = \frac{1}{|G_v^t|} \sum_{\mathbf{I} \in G_v^t} \mathcal{P}_v(\mathbf{I}, c), \quad \bar{\mathcal{S}}_v^t = (1 - \eta) \bar{\mathcal{S}}_v^{t-1} + \eta \mathcal{S}_v^t, \quad (\text{S5})$$

with smoothing factor $\eta = 0.10$. The running score is initialized by the first valid group mean. For compositional prompts during review, the probe evaluates each constituent capability individually, incrementing its assessment count n_v and updating its smoothed score.

Gating, Revocation, and Budget Fallback. Capability states strictly follow the assessment-based definitions in Eq. (2). If a previously mastered capability drops below τ , it leaves $\mathcal{M}$ according to Eq. (2) without a separate margin. An unmastered capability transitions to the fallback set $\mathcal{F}$ once reaching $N_{\max} = 240$ image assessments to prevent curriculum deadlock. Forward progression prioritizes unlocked, unmastered capabilities via Eq. (3), while sampling a 10% replay fraction from $\mathcal{M}$ to maintain acquired competencies. If a prerequisite leaves $\mathcal{M} \cup \mathcal{F}$, downstream successors are temporarily relocked until prerequisites are re-established.

Review and Targeted Remediation. Periodic review is triggered every 3 newly mastered capabilities and runs for 20 rollout iterations, sampling equal proportions (50% single-capability, 50% compositional) from $\mathcal{M} \cup \mathcal{F}$. Capabilities failing the mastery criterion form the remediation set $\mathcal{C}$ as defined in Sec. 4.2. Targeted remediation allocates up to 3 rollout iterations per attempt (maximum 2 retries) to retrain $\mathcal{C}$ under focused prompts before forward progression resumes. Capabilities regaining $\bar{\mathcal{S}}_v \geq \tau$ and $n_v \geq N_{\min}$ re-enter $\mathcal{M}$, while those retaining unmastered status with $n_v \geq N_{\max}$ remain in $\mathcal{F}$.

D.5 TRAINING RESOURCES

Agent-harness alignment uses full-parameter fine-tuning on 16 GPUs. The configured agent-training budget is listed in Tab. S6, and curriculum assessment and review budgets appear in Tab. S9. The case-level normalized latency in Fig. 7 characterizes that example rather than aggregate runtime. Tool parameter inventories are listed in Sec. C.1; they are distinct from the training-resource requirements summarized here.

E EXTENDED RELATED WORK

Reinforcement Learning in Visual Generation. Reinforcement learning (RL) aligns visual generative models through trajectory optimization and preference feedback. For diffusion models, alignment evolved from policy gradients (Black et al., 2024; Fan et al., 2023) and trajectory differentiation (Xu et al., 2023; Clark et al., 2024) to offline preference optimization (Wallace et al., 2024). In

Table S9: Curriculum and capability-assessment hyperparameters.

Parameter	Setting
<i>Capability assessment</i>	
Mastery threshold τ	0.60
Minimum assessed images $N_{\min}$	200
EMA factor η	0.10
Mastery revocation criterion	Evaluated by Eq. (2) ($\bar{S}_v < \tau$); no separate margin
<i>Capability development</i>	
Fallback assessment limit $N_{\max}$	240 assessed images
Assessed images per prompt group	4
Replay probability for mastered capabilities	0.10
<i>Periodic review</i>	
Review interval	Every 3 newly mastered capabilities
Review duration	20 rollout iterations
Single-capability prompt share	50%
Compositional prompt share	50%
<i>Targeted remediation</i>	
Training budget per attempt	3 rollout iterations
Maximum retries per capability	2

flow matching, online policy gradients like Flow-GRPO (Liu et al., 2025a) and DanceGRPO (Xue et al., 2025) enable value-free advantage estimation, complemented by forward matching (Zheng et al., 2026; Xue et al., 2026; Xu et al., 2026b), pairwise advantages (Wang et al., 2025c), and dense stepwise rewards (Deng et al., 2026; Zhang et al., 2026). However, these works rely on well-defined reward and policy learning pipeline, leaving reward hacking and out-of-domain collapse largely un-addressed (Wang et al., 2026c; Tang et al., 2026; Tan et al., 2026). We instead focus on grounded and generalizable reward modeling and rollout organization for progressive capability development.